HW03

1

a

$$t(n) = \begin{cases} 3t(n/3) + \theta(n) & n \geq 2 \\ \theta(1), & n = 1 \end{cases}$$

$$3^k t(n/3^k) + \theta(n)$$

$$k = \log_3 n$$

$$3^{\log_3 n} T(n/3^{\log_3 n}) + k(\theta(n))$$

$$n(T(1)) + \log_3 n(\theta(n)) = n \log n$$

$$\theta(n \log n)$$

b

$$T(n) = \begin{cases} 2T(n/2) + \theta(1), & n \geq 2 \\ \theta(1), & n = 1 \end{cases}$$

$$2^k T(n/2^k) + (\theta(1))$$

$$k = \log_2 n$$

$$n\, T(1) + \theta(1) = n$$

$$\theta(n)$$

c

$$T(n) = \begin{cases} T(n/2) + \theta(n) & n \geq 1 \\ \theta(1), & n = 0 \end{cases}$$

$$T(n/2) = T(n/4) + n/2 + n$$

$$T(n/4) = T(n/8) + [n/4 + n/2 + n]$$

$$T(n/2^k) + n \cdot 2(1 - 1/2^k)$$

$$k = \log_2 n$$

$$T(n/n) + n \cdot 2(1 - 1/n) = 2n - 1 = n$$

$$O(n)$$

2

$\lim_{n\to\infty} f(n)/n = 0$

if first drop $n/2$ the second drop $\theta(n/2)$ worst case

if first drop $\sqrt{n}$ then second drop is $\theta(\sqrt{n})$ worst case

if first drop $\log n$ the second drop $\frac{n}{\log_2} + \log_2$ worst case

best worst case

$\lim_{n\to\infty} \frac{f(\sqrt{n})}{n} = 0$